

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	UP	UPSCALE BURNOUT will force the Input 2 signal to the full scale value when the sensor fails. Diagnostic message IN2 FAIL intermittently flashed on the lower display. The controller remains in Automatic control mode and adjusts the controller output signal in response to the full scale Input 2 signal developed by the Burnout circuitry.
	DOWN	DOWNSCALE BURNOUT will force the Input 2 signal to the lower range value when the sensor fails. Diagnostic message IN2 FAIL intermittently flashed on the lower display. The controller remains in Automatic control mode and adjusts the controller output signal in response to the lower range Input 2 signal developed by the Burnout circuitry.
	NO FS	NO FAILSAFE—This selection does not provide input failure detection and should only be used when a thermocouple input is connected to another instrument which supplies the Burnout current. (For this selection, no burnout signal is sent to the sensor.) ATTENTION For Burnout to function properly on a 0-20 mA input type (or a 0-5V type that uses a dropping resistor), the dropping resistor must be remotely located (across the transmitter terminals). Otherwise, the input at the instrument terminals will always be 0 mA (i.e., within the normal operating range) when the 0-20 mA line is opened.
EMISSIV2	0.01 to 1.00	EMISSIVITY is a correction factor applied to the Radiamatic input signal that is the ratio of the actual energy emitted from the target to the energy which would be emitted if the target were a perfect radiator. Available only for Radiamatic inputs.

3.11 Control Set Up Group

Introduction

The functions listed in this group deal with how the controller will control the process including: Number of Tuning Parameter Sets, Setpoint Source, Tracking, Power-up Recall, Setpoint Limits, Output Direction and Limits, Deadband, and Hysteresis.

Function Prompts

Table 3-11 CONTROL Group Function Prompts

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
PV SOURCE	INPUT 1 INPUT 2 IN ALG1	PROCESS VARIABLE SOURCE —Selects the source of the Process Variable. INPUT 1 INPUT 2 INPUT ALGORITHM 1
PID SETS	1 ONLY 2KEYBD	NUMBER OF TUNING PARAMETER SETS —This selection lets you choose one or two sets of tuning constants (gain, rate, and reset). NOTE: The Tuning Group is automatically configured to have two PID sets when a Duplex Control Algorithm is configured. ONE SET ONLY —Only one set of tuning parameters is available. Configure the values for: Gain (proportional band), Rate, Reset Time, and Cycle Time (if time proportional is used). TWO SETS KEYBOARD SELECTABLE —Two sets of tuning parameters can be configured and can be selected at the operator interface or by using the Digital Inputs.  Press this key until you see PID SET1 or PID SET2 then press ▲ or ▼ to switch between sets. Configure the values for: Gain, Rate, Reset, Cycle Time Gain #2, Rate #2, Reset #2, Cycle #2 Time

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
PID SETS (continued)	2PV SW	TWO SETS PV AUTOMATIC SWITCHOVER—When the process variable is <i>GREATER</i> than the value set at prompt SW VALUE (Switchover Value), the controller will use Gain, Rate, Reset, and Cycle Time. The active PID SET can be read in the lower display. When the process variable is <i>LESS</i> than the value set at prompt SW VALUE, the controller will use Gain #2, Rate #2, Reset #2, and Cycle #2 Time. The active PID SET can be read in the lower display. ATTENTION Other prompts affected: SW VALUE
	2SP SW	TWO SETS SP AUTOMATIC SWITCHOVER—When the setpoint is <i>GREATER</i> than the value set at prompt SW VALUE (Switchover Value), the controller will use Gain, Rate, Reset, and Cycle. When the setpoint is <i>LESS</i> than the value set at prompt SW VALUE, the controller will use Gain #2, Rate #2, Reset #2, and Cycle #2. ATTENTION Other prompts affected: SW VALUE
SW VALUE	Value in engineering units within PV or SP range limits	AUTOMATIC SWITCHOVER VALUE—This is the value of Process Variable or Setpoint at which the controller will switch from Tuning Constant Set #2 to Set #1. ATTENTION Only appears when PID SETS selection is configured for either 2 PVSW or 2 SPSW.
LSP'S	1 ONLY TWO THREE	LOCAL SETPOINT SOURCE—This selection determines what your local setpoint source will be. LOCAL SETPOINT—The setpoint entered from the keyboard. TWO LOCAL SETPOINTS—This selection lets you switch between two local setpoints using the  key. THREE LOCAL SETPOINTS—This selection lets you switch between three local setpoints using the  key

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
RSP SRC	NONE INPUT 2 IN ALG1	REMOTE SETPOINT SOURCE—This selection determines what your remote setpoint source will be when toggled by the SP Select key or Digital Input. NONE—No remote setpoint. INPUT 2—Remote Setpoint is Input 2. IN AL1—Remote Setpoint using Input 1 algorithm. ATTENTION To cycle through the available local setpoints and remote setpoint, press and hold in the  key. When the key is released, the setpoint selection currently displayed will be the new setpoint selection.
AUTOBIAS	DISABLE ENABLE	AUTOBIAS is used for bumpless transfer when transferring from local setpoint to remote setpoint. Auto Bias calculates and adds a bias to remote setpoint input each time a transfer is made. Only available if no tracking is selected. DISABLE—Disables auto bias. ENABLE—Enables auto bias.
SP TRACK	NONE PV RSP	SETPOINT TRACKING—The local setpoint can be configured to track either PV or RSP as listed below. Not configurable when Auto Bias is set. ATTENTION For selections other than NONE, LSP is stored in nonvolatile memory only when there is a mode change; i.e., when switching from RSP to LSP or from Manual to Automatic. If power is lost, then the current LSP value is also lost. NO TRACKING—If local setpoint tracking is not configured, the LSP will not be altered when transfer from RSP to LSP is made. PV—Local setpoint tracks the PV when in manual. RSP—Local setpoint tracks remote setpoint when in automatic. When the controller transfers out of remote setpoint, the last value of the remote setpoint (RSP) is inserted into the local setpoint.
PWR MODE	MANUAL A LSP	POWER UP CONTROLLER MODE RECALL—This selection determines which mode and setpoint the controller will use when the controller restarts after a power loss. MANUAL, LSP—At power-up, the controller will use manual mode with the local setpoint displayed. AUTOMATIC MODE, LAST LSP—At power-up, the controller will use automatic mode with the last local setpoint used before power down displayed.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	A RSP AM SP AM LSP	AUTOMATIC MODE, LAST RSP —At power-up, the controller will use automatic mode with the last remote setpoint used before power down displayed. LAST MODE/LAST SETPOINT used before power down. LAST MODE/LAST LOCAL SETPOINT on power down.
PWR OUT For Three Position Step Control Only (Note 3)	LAST F'SAFE	THREE POSITION CONTROL STEP OUTPUT START-UP MODE —This selection determines what position the motor will be in when powered up or in the failsafe position. LAST OUTPUT —At power-up in automatic mode, the motor position will be the last one prior to power down. When the unit goes into FAILSAFE, it will stay in automatic mode; motor will not be driven to the configured failsafe position. FAILSAFE OUTPUT —At power-up in manual mode, the motor will be driven to either the 0 % or 100 % output position, whichever is selected at prompt FAILSAFE. For Burnout/None, when the unit goes into FAILSAFE, it will go to manual mode; motor will be driven to the configured failsafe position.
SP HiLIM (Note 4)	0 to 100 % of PV span in engineering units	SETPOINT HIGH LIMIT —This selection prevents the local and remote setpoints from going above the value selected here. The setting must be equal or less than the upper range of the PV.
SP LoLIM (Note 4)	0 to 100 % of PV span in engineering units	SET POINT LOW LIMIT —This selection prevents the local and remote setpoints from going below the value selected here. The setting must be equal or greater than the lower range of the PV.
ACTION	DIRECT REVERSE	CONTROL OUTPUT DIRECTION —Select direct or reverse output action. DIRECT ACTING CONTROL —The controller's output <i>increases</i> as the process variable increases. REVERSE ACTING CONTROL —The controller's output <i>decreases</i> as the process variable increases.
OUT RATE	ENABLE DISABLE	OUTPUT CHANGE RATE —Enables or disables the Output Change Rate. The maximum rate is set at prompt PCT/M UP or PCT/M DN. Only available for PID-A, PID-B, PD+MR control algorithms. ENABLE —Allows output rate. DISABLE —Disables output rate.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
PCT/M UP	0 to 9999 % per minute	OUTPUT RATE UP VALUE —This selection limits the rate at which the output can change upward. Enter a value in percent per minute. Appears only if OUT RATE is enabled. “0” means no output rate applied.
PCT/M DN	0 to 9999 % per minute	OUTPUT RATE DOWN VALUE —This selection limits the rate at which the output can change downward. Enter a value in percent per minute. Appears only if OUT RATE is enabled. “0” means no output rate.
OUTHILIM	0 % to 100 % –5 % to 105 %	HIGH OUTPUT LIMIT —This is the highest value of output beyond which you do not want the controller automatic output to exceed. For relay output types. For current output types.
OUTLoLIM	0 % to 100 % –5 % to 105 %	LOW OUTPUT LIMIT —This is the lowest value of output below which you do not want the controller automatic output to exceed. For relay output types. For current output types.
I Hi LIM (Note 5)	Within the range of the output limits	HIGH RESET LIMIT —This is the highest value of output beyond which you do not want reset action to occur
I Lo LIM (Note 5)	Within the range of the output limits	LOW RESET LIMIT —This is the lowest value of output beyond which you do not want reset action to occur.
DROPOFF (Note 5)	–5 to 105 % of output	CONTROLLER DROPOFF VALUE —Output value below which the controller output will drop off to the low output limit value set in prompt OUTLoLIM.
DEADBAND	–5.0 to 25.0 % 0.0 to 25.0 % 0.5 to 5.0 %	DEADBAND is an adjustable gap between the operating ranges of output 1 and output 2 in which neither output operates (positive value) or both outputs operate (negative value). Time Duplex On-Off Duplex Position Proportional and Three Position Step
OUT HYST	0.0 to 100.0 % of PV span	HYSTERESIS (OUTPUT RELAY) is an adjustable overlap of the ON/OFF states of each control output. This is the difference between the value of the process variable at which the control outputs energize and the value at which they de-energize. Only applicable for ON/OFF control.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
FAILMODE	NoLATCH LATCH	FAILSAFE MODE NON LATCHING —Controller stays in last mode that was being used (automatic or manual); If unit was in Automatic mode, then the output goes to the failsafe value. (NOTE 1, NOTE 2) LATCHING —Controller goes to manual mode; If unit was in Automatic mode, then the output goes to the failsafe value. (NOTE 2)
FAILSAFE	0 to 100 %	FAILSAFE OUTPUT VALUE —The value used here will also be the output level when you have Communications SHED set to failsafe or when NO BURNOUT is configured and Input 1 fails. ATTENTION Applies for all output types <i>except</i> Three Position Step Control.
	0 PCT 100 PCT	THREE POSITION STEP FAILSAFE OUTPUT 0 PCT —Motor goes to closed position. 100 PCT —Motor goes to open position.
SW FAIL	0 PCT 100 PCT	Position Proportional motor position when slidewire fails. 0 PCT —Motor goes to closed position. 100 PCT —Motor goes to open position. ATTENTION PWR OUT must be configured for FSAFE.
MAN OUT	0 to 100 %	POWER-UP PRESET MANUAL OUTPUT —At power-up, the controller will go to manual and the output to the value set here. (NOTE 1)
AUTO OUT	0 to 100 %	POWER-UP PRESET AUTOMATIC OUTPUT —At power-up, the controller will begin its automatic control at the output value set here. (NOTE 1)
PBorGAIN	PB PCT	PROPORTIONAL BAND UNITS —Select one of the following for the Proportional (P) term of the PID algorithm: PROPORTIONAL BAND selects units of percent proportional band for the P term of the PID algorithm. <i>Where:</i> $PB \% = \frac{100 \% FS}{GAIN}$

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	GAIN	GAIN selects the unitless term of gain for the P term of the PID algorithm. <i>Where:</i> $GAIN = \frac{100 \% FS}{PB\%}$
MINUTESorRPM	RPM MINUTES	RESET UNITS —Selects units of minutes per repeat or repeats per minute for the I term of the PID algorithm. 20 Repeats per Minute = 0.05 Minutes per Repeat. REPEATS PER MINUTE —The number of times per minute that the proportional action is repeated by reset. MINUTES PER REPEAT —The time between each repeat of the proportional action by reset.
NOTE 1: Does not apply to Three Position Step Control. NOTE 2: If controller is in Manual mode when a failure occurs, then the output will maintain its value. NOTE 3: These selections appear when: A) Control Algorithm is selected for 3PSTEP. B) Control Algorithm is selected for PD+MR and Output Algorithm is selected for Position Proportional. NOTE 4: The local setpoint will automatically adjust itself to be within the setpoint limit range. For example, if SP = 1500 and the SP HiLIM is changed to 1200, the new local setpoint will be 1200. NOTE 5: Reset limits and Dropoff are not displayed when Three Position Step Control is configured.		

Introduction

Function Prompts

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
AUX OUT ATTENTION Prompts for the Auxiliary Output Selection appear only if one of the Auxiliary Output boards is installed.		AUXILIARY OUTPUT SELECTION This selection provides an mA output representing one of several control parameters. The display for auxiliary output viewing will be in engineering units for all but output. Output will be displayed in percent. ATTENTION <i>Other prompts affected by these selections: 4mA VAL and 20mA VAL.</i> ATTENTION Output cannot be configured when Three Position Step Control is used.
	DISABLE	NO AUXILIARY OUTPUT
	INPUT 1	INPUT 1—This represents the configured range of input 1. FOR EXAMPLE: Type J Thermocouple (0 °F to 1600 °F) 0 °F display = 0 % output 1600 °F display = 100 % output
	INPUT 2	INPUT 2 represents the value of the configured range of input 2.
	PV	PROCESS VARIABLE—Represents the value of the Process Variable. $PV = \text{Input } X \times \text{Ratio } X + \text{Bias } X$

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	DEV	DEVIATION (PROCESS VARIABLE MINUS SETPOINT)—Represents –100 % to +100 % of the selected PV span in engineering units. Zero deviation will produce a center scale (12 mA or 50 %) output. A negative deviation equal in magnitude to the Auxiliary Output High Scaling Factor will produce a low end output (4 mA or 0 %) output. A positive deviation equal in magnitude to the Auxiliary Output Low Scaling Factor will produce a high end output (20 mA or 100 %). FOR EXAMPLE: Input 1 = Type T High Thermocouple PV range = –300 °F to +700 °F PV span = 1000 °F Deviation Range = –1000 °F to +1000 °F Auxiliary Output Low Scale Value = 0.0 Auxiliary Output High Scale Value = 1000 If PV = 500 °F and SP = 650 °F then Deviation Display = –150 °F, which is –7.5% of the Deviation Range, so Auxiliary Output = 50% – 7.5% = 42.5%
	OUTPUT	OUTPUT—Represents the displayed controller output in percent (%). Cannot be used with Three Position Step Control.
	SP	SETPOINT—Represents the value of the setpoint currently in use (LSP1, LSP2, LSP3, RSP or CSP) and is shown in the same units as those used by the PV.
	LSP 1	LOCAL SETPOINT ONE—Auxiliary output represents Local Setpoint 1 regardless of active setpoint.
	RSP	REMOTE SETPOINT – Represents the configured RSP regardless of the active SetPoint.
	IN ALG1	INPUT ALGORITHM 1 OUTPUT—Represents the output from input algorithm 1.
CO RANGE	4-20mA 0-20mA	AUXILIARY CURRENT OUTPUT RANGE—Allows the user to easily select 4-20mA output or 0-20mA output operation without the need for recalibration of the instrument. ATTENTION Changing the Auxiliary Current Output Range will result in the loss of Field Calibration values and will restore Factory Calibration values.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
LOW VAL	Low Scale Value within the range of the selected variable to represent the minimum output (0 or 4 mA)	AUXILIARY OUTPUT LOW SCALING FACTOR—This is a value in engineering units used to represent all AUX OUT parameters except Output. For Output, this is a value in percent and can be any value between –5 % and +105 %. However, keep in mind that relay output types can only be scaled 0 % to 100 %.
HIGH VAL	High Scale Value within the range of the selected variable to represent the maximum output (20 mA)	AUXILIARY OUTPUT HIGH SCALING FACTOR—This is a value in engineering units used to represent all AUX OUT parameters except Output. For Output, this is a value in percent and can be any value between –5 % and +105 %. However, keep in mind that relay output types can only be scaled 0 % to 100 %.
DIG INP1	NONE TO MAN TO LSP TO 2SP TO 3SP TO DIR	DIGITAL INPUT 1 SELECTIONS—All selections are available for Input 1. The controller returns to its original state when contact opens, except when overruled by the keyboard. NO DIGITAL INPUT SELECTIONS TO MANUAL—Contact closure puts the affected loop into manual mode. Contact open returns controller to former mode. TO LOCAL SETPOINT—When a remote setpoint is configured, contact closure puts the controller into local setpoint 1. When contact opens, the controller returns to former operation—local or remote setpoint—unless SP Select key is pressed while digital input is active. If this happens, the controller will stay in the local setpoint mode when contact opens. TO LOCAL SETPOINT TWO—Contact closure puts the controller into local setpoint 2. TO LOCAL SETPOINT THREE—Contact closure puts the controller into local setpoint 3. TO DIRECT ACTION—Contact closure selects direct controller action.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	TO HOLD	TO HOLD—Contact closure suspends Setpoint Program or Setpoint Ramp. When contact reopens, the controller starts from the Hold point of the Ramp/Program unless the Ramp/Program was not previously started via the  key. This selection applies to either loop.
	TO PID2	TO PID2 —Contact closure selects PID Set 2.
	PV 2IN	PV=INPUT 2 —Contact closure selects PV = Input 2.
	RERUN	RERUN --Allows the Setpoint Programmer to be reset to the initial segment of its current cycle, unit stays in previous mode.
	TO RUN	RUN—Contact closure starts a stopped SP Ramp or Program. Upper left character blinks "R". Reopening the contact puts controller in HOLD mode. This selection applies to either loop.
	ToBEGIN	EXTERNAL SP PROGRAM RESET—Contact closure resets SP Program back to the beginning of the first segment in the program and places the program in the HOLD mode. Program cycle number is not affected. Reopening switch has no effect. This selection applies to either loop. ATTENTION Once the last segment of the setpoint program has timed out, the controller enters the mode of action specified in the configuration data and the program cannot be reset to the beginning of the first segment by digital input closure.
	STOP I	INHIBIT INTEGRAL (RESET) —Contact closure disables PID Integral (Reset) action.
	MAN FS	MANUAL FAILSAFE OUTPUT—Controller goes to Manual mode, output goes to the Failsafe value. ATTENTION This will cause a bump in the output when switching from Automatic to Manual. The switch back from Manual to Automatic is bumpless. When the switch is closed, the output can be adjusted from the keyboard.
	TO LOCK	KEYBOARD LOCKOUT —Contact closure disables all keys. Lower display shows LOCKED if a key is pressed.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	TO Aout	AUTOMATIC OUTPUT —Contact closure sends output to the value set at Control prompt AUTO OUT when the controller is in the Automatic mode. Reopening the contact returns the controller to the normal output. ATTENTION Does not apply to Three Position Step Control.
	TIMER	TIMER —Contact closure starts timer, if enabled. Reopening the switch has no effect.
	AM STA	TO AUTO/MANUAL STATION —Contact closure causes the control loop to perform as follows: PV = Input 2 Action = Direct Control algorithm = PD+MR PID SET = 2 SP = LSP 2
	TO TUNE	INITIATE LIMIT CYCLE TUNING —Contact closure starts the tuning process. The lower display shows TUNE ON. Opening the contact has no effect.
	SP Init	SETPOINT INITIALIZATION —Contact closure forces the setpoint to the current PV value. Opening the contact has no effect.
	TRACK	OUTPUT TRACKS INPUT 2 —Contact closure allows Output to track Input 2. While the switch is open, the output is in accordance with its pre-defined functionality. When the switch is closed, the output value (in percent) will track the Input 2 percent of range value. When the switch is reopened, the output will start at this last output value and normal PID action will then take over control. The transfer is bumpless.
	TO RSP	TO REMOTE SETPOINT —Contact closure selects the Remote setpoint.
	RST FB	EXTERNAL RESET FEEDBACK —Contact closure allows Input 2 to override the internal reset value.
	To PURGE	TO PURGE —Contact closure forces the loop to Manual mode with the output set to the Output High Limit configuration. MAN lights and the Output value is shown on the lower display. Opening the switch has no effect. ATTENTION Does not apply to Three Position Step Control.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	Lo FIRE MAN LAT PV Hold	LOW FIRE—Contact closure forces the loop to Manual mode with the output set to the Output Low Limit configuration. MAN lights and the Output value is shown on the lower display. Opening the switch has no effect. ATTENTION Does not apply to Three Position Step Control. MANUAL LATCHING—Contact closure transition forces the loop to Manual mode. Opening the switch has no effect. If the MAN/AUTO key is pressed while the switch is closed, the loop will return to Automatic mode. PROCESS VARIABLE HOLD—when the switch is closed, PV is frozen at last value. When switch opens, PV resumes.
DIG 1COMB	DISABLE +PID2 +TO DIR +TO SP2 +DIS AT +TO SP1 +RUN	DIGITAL INPUT 1 COMBINATION SELECTIONS—This selection allows the specified function to occur in addition to the one chosen for DIG IN 1. DISABLE—Disables combination function. PLUS PID2—Contact closure selects PID Set 2. PLUS DIRECT ACTION—Contact closure selects direct controller action. PLUS SETPOINT 2—Contact closure puts the controller into setpoint 2. PLUS DISABLE ADAPTIVE TUNE—Contact closure disables Accutune process. PLUS SETPOINT 1—Contact closure puts the controller into setpoint 1. PLUS RUN SETPOINT PROGRAM/RAMP—Contact closure starts SP Program/Ramp if enabled.
DIG INP2	Same selections as for Digital Input 1	DIGITAL INPUT 2 SELECTIONS
DIG2COMB	Same selections as Digital Input 1 Combinations	DIGITAL INPUT 2 COMBINATIONS

3.13 Communications Group

Introduction

The Communications group lets you configure the controller to be connected to a host computer via Modbus® or Ethernet TCP/IP protocol.

Introduction

A controller with a communications option looks for messages from the host computer. If these messages are not received within the configured shed time, the controller will SHED from the communications link and return to stand-alone operation. You can also set the SHED output mode and setpoint recall, and communication units.

Up to 99 addresses can be configured over this link. The number of units that can be configured depends on the link length, with 31 being the maximum for short link lengths and 15 drops being the maximum at the maximum link length.

Function Prompts

Table 3-13 Communications Group Function Prompts

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
Com ADDR	1 to 99	COMMUNICATIONS STATION ADDRESS —This is a number that is assigned to a controller that is to be used with the communications option. This number will be its address.
ComSTATE	DISABLE MODBUS	COMMUNICATIONS SELECTION DISABLE —Disables RS-485 communications option. MODBUS —Enables RS-485 Modbus RTU communication port.
IR ENABLE	DISABLE ENABLE	IR ENABLE —Disable/Enables IR communications port. ATTENTION If there are no IR communications transactions for four minutes, then the IR port automatically shuts down. It can be re-enabled by pressing any key on the front panel.
BAUD	4800 9600 19200 38400	BAUD RATE is the transmission speed in bits per second. This value is used for both RS-485 and IR Communications, but for IR Communications, values below 19200 baud are interpreted as being 19200 baud. 4800 BAUD 9600 BAUD 19200 BAUD 38400 BAUD

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition																								
TX DELAY	1 to 500 milliseconds	TX DELAY —Configurable response-delay timer allows you to force the instrument to delay its response for a time period of from 1 to 500 milliseconds compatible with the host system hardware/software.																								
WS FLOAT	FP_B FP_BB FP_L FP_LB	Defines word/byte order of floating point data for communications. Byte values: <table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>seeeeeee</td><td>emmmmmmm</td><td>mmmmmmmm</td><td>mmmmmmmm</td></tr></table> Where: s = sign, e = exponent, m = mantissa bit <table><tr><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>1</td><td>0</td><td>3</td><td>2</td></tr><tr><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>2</td><td>3</td><td>0</td><td>1</td></tr></table>	0	1	2	3	seeeeeee	emmmmmmm	mmmmmmmm	mmmmmmmm	0	1	2	3	1	0	3	2	3	2	1	0	2	3	0	1
0	1	2	3																							
seeeeeee	emmmmmmm	mmmmmmmm	mmmmmmmm																							
0	1	2	3																							
1	0	3	2																							
3	2	1	0																							
2	3	0	1																							
SHED ENAB	DISABLE ENABLE	SHED ENABLE —Disables/enables shed functionality. Applies to Modbus protocol only.																								
SHEDTIME	0 to 255	SHED TIME —The number that represents how many sample periods there will be before the controller sheds from communications. Each period equals 1/3 seconds; 0 equals No shed. Note: If ComSTATE is set to MODBUS or MB3K and if SHEDENAB is set to DISABL, Shed Time will not be configurable.																								
SHEDMODE	LAST TO MAN FSAFE TO AUTO	SHED CONTROLLER MODE AND OUTPUT LEVEL—Determines the mode of local control you want when the controller is shed from the communications link. LAST—SAME MODE AND OUTPUT—The controller will return to the same mode (manual or automatic) at the same output level that it had before shed. TO MAN—MANUAL MODE, SAME OUTPUT—The controller will return to manual mode at the same output level that it had before shed. FSAFE—MANUAL MODE, FAILSAFE OUTPUT—The controller will return to manual mode at the output value selected at Control prompt FAILSAFE. TO AUTO—AUTOMATIC MODE, LAST SP—The controller will return to the automatic mode and the last setpoint used before shed.																								

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
SHED SP	TO LSP TO CSP	SHED SETPOINT RECALL Note: If SHEDENAB=DISABLE, this prompt will not be configurable. TO LSP—Controller will use last local or remote setpoint used. TO CSP—When in “slave” mode, the controller will store the last host computer setpoint and use it at the Local setpoint. When in “monitor” mode, the controller will shed to the last instrument Local or Remote setpoint used, and the LSP is unchanged.
UNITS	ENG PERCENT	COMPUTER SETPOINT UNITS ENG — Engineering units PERCENT — Percent of PV range
CSP RATO	–20.0 to 20.0	COMPUTER SETPOINT RATIO —Computer setpoint ratio.
CSP BIAS	–999. to 9999. (engineering units)	COMPUTER SETPOINT BIAS —Computer setpoint bias in Engineering Units.
LOOPBACK	DISABLE ENABLE	LOCAL LOOPBACK tests the RS-485 communications port. It is not used for any other communications port. DISABLE—Disables the Loopback test. ENABLE—Allows loopback test. The instrument goes into Loopback mode in which it sends and receives its own message. The instrument displays PASS or FAIL status in the upper display and LOOPBACK in the lower display while the test is running. The instrument will go into manual mode when LOOPBACK is enabled with the output at the Failsafe value. The test will run until the operator disables it here, or until power is turned off and on. ATTENTION The instrument does not have to be connected to the external communications link in order to perform this test. If it is connected, only one instrument should run the loopback test at a time. The host computer should not be transmitting on the link while the loopback test is active.

3.14 Alarms Set Up Group

Introduction

An alarm is an indication that an event that you have configured (for example—Process Variable) has exceeded one or more alarm limits. There are two alarms available. Each alarm has two setpoints. You can configure each of these two setpoints to alarm on various controller parameters.

There are two alarm output selections, High and Low. You can configure each setpoint to alarm either High or Low. These are called single alarms.

You can also configure the two setpoints to alarm on the same event and to alarm both high and low. A single adjustable Hysteresis of 0 % to 100 % is configurable for the alarm setpoint.

See Table 2-3 in the *Installation* section for Alarm relay contact information.

The prompts for the Alarm Outputs appear whether or not the alarm relays are physically present. This allows the Alarm status to be shown on the display and/or sent via communications to a host computer.

Function Prompts

Table 3-14 ALARMS Group Function Prompts

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
A1S1 VAL	Value in engineering units	ALARM 1 SETPOINT 1 VALUE —This is the value at which you want the alarm type chosen in prompt A1S1TYPE to actuate. The value depends on what the setpoint has been configured to represent. No setpoint is required for alarms configured for Communications SHED. For SP Programming the value is the segment number for which the event applies. This prompt does not appear for “Alarm on Manual” type alarm. For example: A1S1TYPE = MANUAL.
A1S2 VAL	Value in engineering units	ALARM 1 SETPOINT 2 VALUE —This is the value at which you want the alarm type chosen in prompt A1S2TYPE to actuate. The details are the same as A1S1 VAL.
A2S1 VAL	Value in engineering units	ALARM 2 SETPOINT 1 VALUE —This is the value at which you want the alarm type chosen in prompt A2S1TYPE to actuate. The details are the same as A1S1 VAL.
A2S2 VAL	Value in engineering units	ALARM 2 SETPOINT 2 VALUE —This is the value at which you want the alarm type chosen in prompt A2S2TYPE to actuate. The details are the same as A1S1 VAL.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
A1S1TYPE	NONE INPUT 1 INPUT 2 PV DEV OUTPUT SHED EV ON EV OFF MANUAL REM SP F SAFE PV RATE DIG INP 1 DIG INP 2 DEV 2 BREAK TCWARN TCFail PVHOLD	ALARM 1 SETPOINT 1 TYPE—Select what you want Setpoint 1 of Alarm 1 to represent. It can represent the Process Variable, Deviation, Input 1, Input 2, Output, and if you have a model with communications, you can configure the controller to alarm on SHED. If you have setpoint programming, you can alarm when a segment goes ON or OFF. NO ALARM INPUT 1 INPUT 2 PROCESS VARIABLE DEVIATION OUTPUT (NOTE 1) SHED FROM COMMUNICATIONS EVENT ON (SP PROGRAMMING) EVENT OFF (SP PROGRAMMING) ALARM ON MANUAL MODE (NOTE 2) REMOTE SETPOINT FAILSAFE PV RATE OF CHANGE DIGITAL INPUT 1 ACTUATED (NOTE 7) DIGITAL INPUT 2 ACTUATED (NOTE 7) DEVIATION FROM LSP 2 (NOTE 3) LOOP BREAK (NOTE 4) THERMOCOUPLE WARNING (NOTE 5) THERMOCOUPLE FAIL (NOTE 6) PV HOLD ATTENTION NOTE 1. When the controller is configured for Three Position Step Control, alarms set for Output will not function. NOTE 2. Alarm 1 is not available if the Timer is enabled because Alarm 1 is dedicated to Timer output. NOTE 3. This Deviation Alarm is based upon deviation from the 2nd Local Setpoint or Remote SP regardless of whichever SP is active. NOTE 4. Loop Break monitors the control loop to determine if it is working. When enabled, the control output is checked against the minimum and maximum output limit settings. When the output reaches one of these limits, a timer begins. If the timer expires and the output has not caused the PV to move by a pre-determined amount, then the alarm activates, thus signalling that the loop is broken. The loop break timer value must be configured by the operator as the AxSx VAL entry. This value is in seconds with a range of 0 to 3600 seconds. A setting of 0 is equivalent to an instantaneous loop break when the output reaches one of its limit

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
		values. NOTE 5. Thermocouple Warning means that the instrument has detected that the Thermocouple Input is starting to fail. Not valid for other input types. NOTE 6. Thermocouple Failing means that the instrument has detected that the Thermocouple Input is in imminent danger of failing. Not valid for other input types. NOTE 7: For the Digital Input selections, DIG INP1 can be either enabled or disabled in the Options Group (See <i>Sub-Section 3.12</i>), but DIG INP2 must be enabled in the Options Group for the alarm to function properly.
A1S1 H L	HIGH LOW	<i>If Setpoint Programming is disabled or if the Alarm Type is not configured for Event On/Off:</i> ALARM 1 SETPOINT 1 STATE—Select whether you want the alarm type chosen in prompt A1S1TYPE to alarm High or Low. HIGH ALARM LOW ALARM <i>If Setpoint Programming is enabled and if the Alarm Type is configured for Event On/Off:</i> ALARM 1 SEGMENT EVENT 1—Select whether you want the alarm type chosen in prompt A1S1TYPE to alarm the beginning or end of a segment in setpoint Ramp/Soak programming. BEGINNING OF SEGMENT END OF SEGMENT ATTENTION Alarms configured for events will not operate on Setpoint Program segments of zero length.
A1S1 EV	BEGIN END	

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
A1S2TYPE	Same as A1S1 TYPE	ALARM 1 SETPOINT 2 TYPE —Select what you want Setpoint 2 of Alarm 1 to represent. The selections are the same as A1S1TYPE.
A1S2 H L A1S2 EV	HIGH LOW BEGIN END	ALARM 1 SETPOINT 2 STATE —Same as A1S1 H L. ALARM 1 SEGMENT EVENT 2 —Same as A1S1 EV.
A2S1TYPE	Same as A1S1 TYPE	ALARM 2 SETPOINT 1 TYPE —Select what you want Setpoint 1 of Alarm 2 to represent. The selections are the same as A1S1TYPE. ATTENTION Not available with Relay Duplex or Position Proportional output types unless using Dual Relay PWA.
A2S1 H L A2S1 EV	HIGH LOW BEGIN END	ALARM 2 SETPOINT 1 STATE —Same as A1S1 H L. ALARM 2 SEGMENT EVENT 1 —Same as A1S1 EV.
A2S2TYPE	Same as A1S1 TYPE	ALARM 2 SETPOINT 2 TYPE —Select what you want Setpoint 2 of Alarm 2 to represent. The selections are the same as A1S1TYPE. ATTENTION Not applicable with Relay Duplex or Position Proportional output types unless using Dual Relay PWA.
A2S2 H L A2S2 EV	HIGH LOW BEGIN END	ALARM 2 SETPOINT 2 STATE —Same as A1S1 H L. ALARM 2 SEGMENT EVENT 2 —Same as A1S1 EV.
ALHYST	0.0 to 100.0 % of span or full output as appropriate	ALARM HYSTERESIS —A single adjustable hysteresis is provided on alarms such that when the alarm is OFF it activates at exactly the alarm setpoint; when the alarm is ON, it will not deactivate until the variable is 0.0 % to 100 % away from the alarm setpoint. Configure the hysteresis of the alarms based on INPUT signals as a % of input range span. Configure the hysteresis of the alarm based on OUTPUT signals as a % of the full scale output range.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
ALM OUT1	NoLATCH LATCH	LATCHING ALARM OUTPUT 1—Alarm output 1 can be configured to be Latching or Non-latching. NoLATCH —Non-latching LATCH—Latching ATTENTION When configured for latching, the alarm will stay active after the alarm condition ends until the RUN/HOLD key is pressed.
BLOCK	DISABLE ALARM 1 ALARM 2 ALARM12	ALARM BLOCKING—Prevents nuisance alarms when the controller is first powered up. The alarm is suppressed until the parameter gets to the non-alarm limit or band. Alarm blocking affects both alarm setpoints. DISABLE—Disables blocking ALARM 1—Blocks alarm 1 only ALARM 2—Blocks alarm 2 only ALARM12—Blocks both alarms ATTENTION When enabled on power up or initial enabling via configuration, the alarm will not activate unless the parameter being monitored has not been in an alarm condition for a minimum of one control cycle (167 ms).
DIAGNOST	DISABLE ALARM 1 ALARM 2	DIAGNOSTIC—Monitors the Current Output and/or Auxiliary Output for an open circuit condition. If either of these two outputs falls below about 3.5 mA, then an Alarm is activated. This configuration is in addition to whatever was selected for AxSxTYPE. DISABLE—Disables Diagnostic Alarm ALARM 1—Alarm 1 is diagnostic alarm ALARM 2—Alarm 2 is diagnostic alarm

3.15 Display Set Up Group

Introduction

This group includes selections for Decimal place, Units of temperature, Language and Power frequency.

Function Prompts

Table 3-15 DISPLAY Group Function Prompts

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
DECIMAL	NONE ONE TWO THREE	DECIMAL POINT LOCATION—This selection determines where the decimal point appears in the display. NONE—No Decimal Place—fixed, no auto-ranging ONE—One Place TWO—Two Places THREE—Three Places ATTENTION Auto-ranging will occur for selections of one, two or three decimal places. For example, should the instrument be configured for two decimal places and the PV exceeds 99.99, then the display will change to a single decimal place so that values of 100.0 and above can be shown.
TEMP UNIT	DEG F DEG C NONE	TEMPERATURE UNITS—This selection will affect the indication and operation. DEG F—Degrees Fahrenheit – Degrees F Annunciator lighted DEG C—Degrees Centigrade – Degrees C Annunciator lighted NONE—No temperature annunciators lighted. Upper and Lower Displays will show temperature in Degrees Fahrenheit when inputs are configured for Thermocouple or RTD types.
PWR FREQ	60 HZ 50 HZ	POWER LINE FREQUENCY—Select whether your controller is operating at 50 or 60 Hertz. ATTENTION For controllers powered by +24 Vdc, this configuration should be set to the AC line frequency used to produce the +24 Vdc supply. Incorrect setting of this parameter may cause normal mode noise problems in the input readings.
RATIO 2	DISABLE	INPUT 2 RATIO—This enables the Ratio for Input 2 to be set from the front panel. Input 2 must be installed and enabled for this configuration to operate. DISABLE—Disables setting Ratio 2 from front panel.

Function Prompt Lower Display	Selections or Range of Setting Upper Display	Parameter Definition
	ENABLE	ENABLE —Allows the Ratio for Input 2 to be set through the keyboard.
LANGUAGE	ENGLISH FRENCH GERMAN SPANISH ITALIAN	LANGUAGE —This selection designates the prompt language. ENGLISH FRENCH GERMAN SPANISH ITALIAN

3.16 Configuration Record Sheet

Enter the value or selection for each prompt on this sheet so you will have a record of how your controller was configured.

Group Prompt	Function Prompt	Value or Selection	Factory Setting	Group Prompt	Function Prompt	Value or Selection	Factory Setting
TUNING	PROP BD	_____	--	ALGORITHM	CONT ALG	_____	PID A
	or				TIMER	_____	DISABLE
	GAIN	_____	1.000		PERIOD	_____	0.01
	or				START	_____	KEY
	GAINVALn	Read Only	--		LOW DISP	_____	TI REM
	RATE MIN	_____	0.00		RESET	_____	KEY
	RSET MIN	_____	1.00		INCREMENT	_____	MINUTES
	or				INP ALG1	_____	NONE
	RSET RPM	_____	--		MATH K	_____	1.0
	MAN RSET	_____	0		CALC HI	_____	--
	PROPB2	_____	--		CALC LO	_____	--
	or			OUT ALG	ALG1 INA	_____	INPUT 1
	GAIN 2	_____	1.000		ALG1 INB	_____	INPUT 2
	RATE2MIN	_____	0.00		ALG1 INC	_____	NONE
	RSET2MIN	_____	1.00		ALG1BIAS	_____	0.000
	or				PCT CO	_____	0.200
	RSET2RPM	_____	--				
	CYC SEC	_____	20		OUT ALG	_____	NOTE 1
	or				RLYSTATE	_____	1OF 2ON
	CYC SX3	_____	20		RLY TYPE	_____	MECHAN
	CYC2 SEC	_____	20		MOTOR TI	_____	30
SP RAMP	or				CUR OUT	_____	DISABLE
	CYC2 SX3	_____	20		CO RANGE	_____	4-20mA
	SECURITY	_____	0		LOW VAL	_____	0.0
	LOCKOUT	_____	CALIB		HIGH VAL	_____	100.0
	AUTO MAN	_____	ENABLE				
	RUN HOLD	_____	ENABLE				
	SP SEL	_____	ENABLE				
ACCUTUNE	SP RAMP	_____	DISABLE	INPUT 1	IN1 TYPE	_____	0-10mV
	TIME MIN	_____	3		XMITTER1	_____	LINEAR
	FINAL SP	_____	1000		IN1 HIGH	_____	1000
	SP RATE	_____	DISABLE		IN1 LOW	_____	0
	EU/HR UP	_____	0		RATIO 1	_____	1.00
	EU/HR DN	_____	0		BIAS IN1	_____	0
	HOTSTART	_____	DISABLE		FILTER 1	_____	1
	SP PROG	_____	DISABLE		BURNOUT1	_____	NONE
ACCUTUNE					EMMISIV1	_____	0.00
	FUZZY	_____	DISABLE	INPUT 2	IN2 TYPE	_____	0-10mV
	ACCUTUNE	_____	DISABLE		XMITTER2	_____	LINEAR
	DUPLEX	_____	MANUAL		IN2 HIGH	_____	1000
	AT ERROR	Read Only	NONE		IN2 LOW	_____	0
					RATIO 2	_____	1.00
					BIAS IN2	_____	0
					FILTER 2	_____	1
					BURNOUT2	_____	NONE
					EMMISIV2	_____	0.00

NOTE 1: Model Number dependent.

Group Prompt	Function Prompt	Value or Selection	Factory Setting	Group Prompt	Function Prompt	Value or Selection	Factory Setting
CONTROL	PV SOURC	_____	INPUT 1	ALARMS	A1S1TYPE	_____	NONE
	PID SETS	_____	1 ONLY		A1S1 VAL	_____	90
	SW VALUE	_____	0.00		A1S1 H L	_____	HIGH
	LSP'S	_____	1 ONLY		A1S1 EV	_____	--
	RSP SRC	_____	NONE		A1S2TYPE	_____	NONE
	AUTOBIAS	_____	DISABLE		A1S2 VAL	_____	10
	SP TRACK	_____	NONE		A1S2 H L	_____	LOW
	PWR MODE	_____	MANUAL		A1S2 EV	_____	--
	PWR OUT	_____	LAST		A2S1TYPE	_____	NONE
	SP HiLIM	_____	1000		A2S1 VAL	_____	95
	SP LoLIM	_____	0		A2S1 H L	_____	HIGH
	ACTION	_____	REVERSE		A2S1 EV	_____	--
	OUT RATE	_____	DISABLE		A2S2TYPE	_____	NONE
	PCT/M UP	_____	0		A2S2 VAL	_____	5
	PCT/M DN	_____	0		A2S2 H L	_____	LOW
	OUTHILIM	_____	100		A2S2 EV	_____	--
	OUTLoLIM	_____	0.0		AL HYST	_____	0.1
	I Hi LIM	_____	100.0		ALM OUT1	_____	NoLATCH
	I Lo LIM	_____	0.0		BLOCK	_____	DISABLE
	DROPOFF	_____	0		DIAGNOST	_____	DISABLE
	DEADBAND	_____	1.0	DISPLAY	DECIMAL	_____	NONE
	OUT HYST	_____	0.5		TEMPUNIT	_____	NONE
	FAILMODE	_____	NoLATCH		PWR FREQ	_____	60 HZ
	FAILSAFE	_____	0.0		RATIO 2	_____	DISABLE
	MAN OUT	_____	0.0		LANGUAGE	_____	ENGLISH
	AUTO OUT	_____	0.0				
	PBorGAIN	_____	GAIN				
	MINorRPM	_____	MIN				
OPTIONS	AUX OUT	_____	DISABLE	Ethernet (Accessible via PIE Tool)	MAC Add.	_____	--
	CO RANGE	_____	4-20mA		IP Address	_____	10.0.0.2
	LOW VAL	_____	0.0		Subnet Mask	_____	225.225.225.0
	HIGH VAL	_____	100.0		Default Gate	_____	0.0.0.0
	DIG INP1	_____	NONE		To Email	_____	--
	DIG1COMB	_____	DISABLE		SMTP Add.	_____	0.0.0.0
	DIG INP2	_____	NONE		Alarm	_____	NONE
	DIG2COMB	_____	DISABLE		Email Subj	_____	--
COM	Com ADDR	_____	3				
	ComSTATE	_____	DISABLE				
	IR ENABLE	_____	ENABLE				
	BAUD	_____	19200				
	TX DELAY	_____	1				
	WS FLOAT	_____	FP B				
	SHEDENAB	_____	DISABLE				
	SHEDTIME	_____	30.0				
	SHEDMODE	_____	LAST				
	SHEDSP	_____	TO LSP				
	UNITS	_____	PERCNT				
	CSP RATO	_____	1.0				
	CSP BIAS	_____	0				
	LOOPBACK	_____	DISABLE				